

Workshop Schedule

Time	Topic/activity	Instructor
9:00-9:10	Introductions and workshop overview	All
9:10-9:30	Introduction of conceptual ideas	Emily Palmquist
9:30-10:10	Introduction to probability, Bayes rule, and “basic” Bayes	Jessica Guo
10:10-10:50	Introduction to hierarchical Bayesian modeling	Kiona Ogle
10:50-11:00	Break	
11:00-12:00	Continuation of hierarchical Bayesian modeling	Kiona Ogle
12:00-1:00	Lunch	
1:00-2:10	JAGS demonstration of a basic Bayesian problem; code, implementation, convergence, mixing, posterior stats	Alicia Formanack
2:10-2:20	Brief overview of datasets/problems for group projects	Alicia Formanack
2:20-3:50	Working groups (form groups of ~ 4 participants each; work through chosen project)	All
3:50-4:00	Closing remarks and “where to go from here”	



Why use Bayesian modeling in ecology?

Ecological Society of America Annual Meeting
26 July 2026

Emily Palmquist | *U.S. Geological Survey*
Jessica Guo | *Harvey Mudd College*
Kiona Ogle | *Northern Arizona University*
Alicia Formanack | *Northern Arizona University*
Eva Deegan | *Northern Arizona University*

Photos (left to right):
USGS
E. Palmquist
Freshwater Illustrated

Why Bayesian? - Interpretation

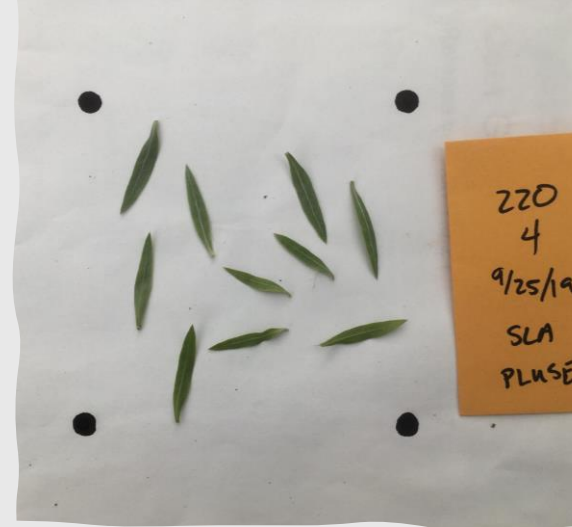
- Emphasis placed on **estimation** rather than hypothesis testing
- Direct quantification of **uncertainty**
- Bayesian credible interval is the **probability that an unknown parameter lies within a specific range** based on observed data
 - Vs. a confidence interval - the resulting interval would include the true parameter 95% of the time if the experiment was repeated over and over

Why Bayesian? - Practical

- Specify model to fit the data rather than the other way around
 - Treats data as drawn from a distribution, any distribution
- Flexible, can accommodate ecological research
 - Unbalanced designs
 - Missing data
 - Non-linear and non-Gaussian processes
 - Non-Gaussian processes
 - Fixed and random effects, nested or not
 - Multiple sources of uncertainty (including latent processes)
- Combines information
 - Related datasets can be modeled in a single analysis
 - Prior knowledge can be included as informative priors

Complicated experimental design?

- Levels nested within buckets within pumped systems
- Gene x environment interactions
- Height, biomass, growth, photosynthesis, etc. on one individual is all related

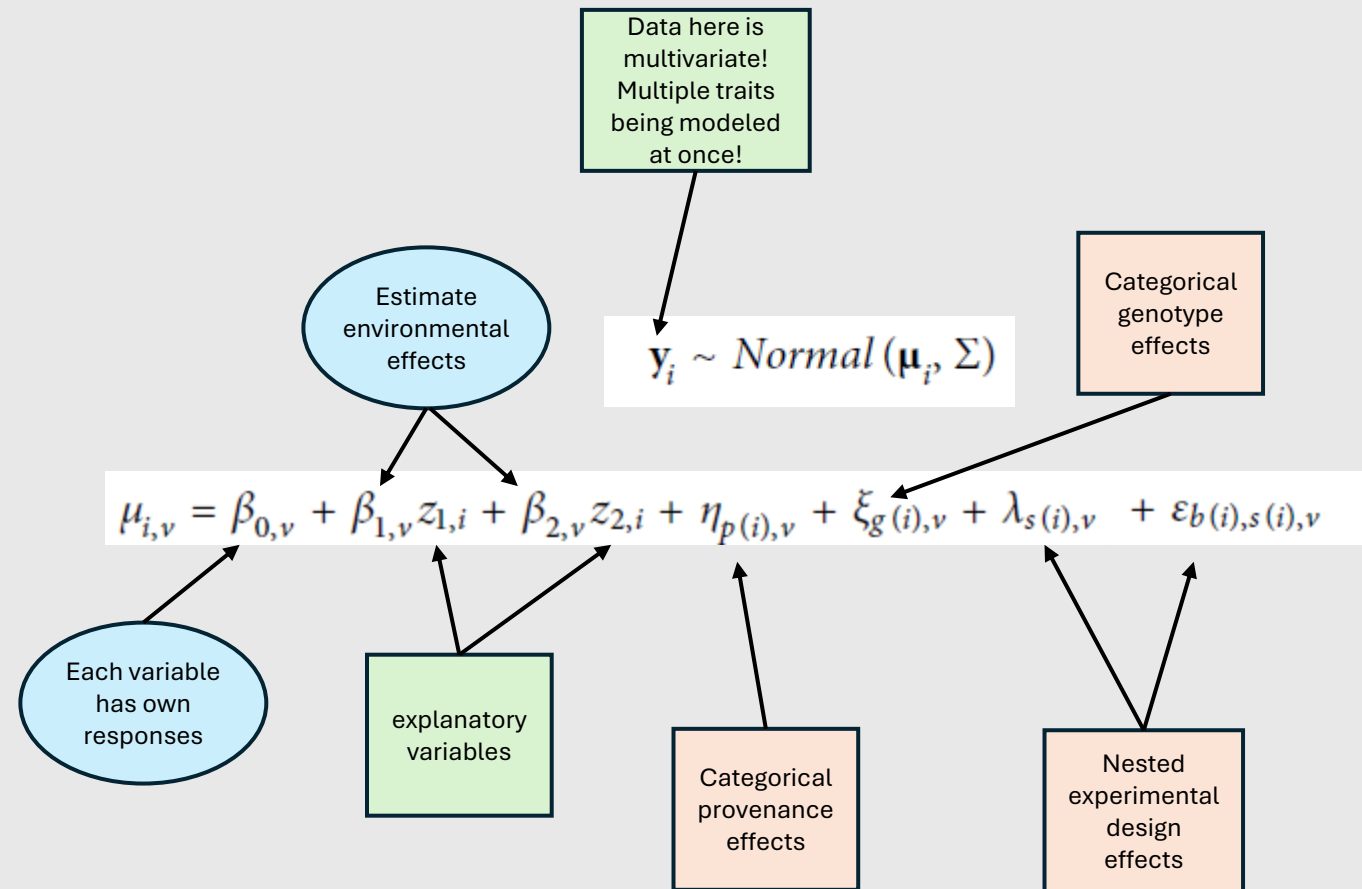


Complicated experimental design? – No Problem

- Levels nested within buckets within pumped systems
- Gene x environment interactions
- Height, biomass, growth, photosynthesis, etc. on one individual is all related

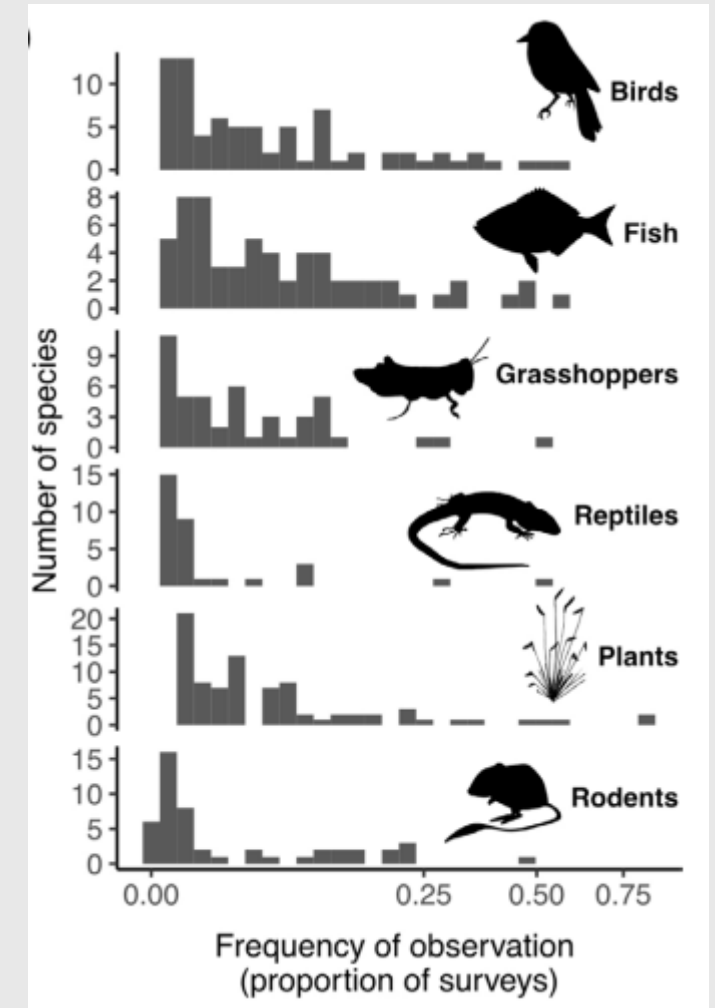
Solutions

- Hierarchical model structure
- Random and fixed effects, crossed or not
- Avoid non-identifiability when using multiple random effects
- Multivariate model for linked variables



Quantify detection error?

- Trying to correlate imperfect observation data to environmental data
- How accurate are your multispecies occupancy-abundance models ?
- Scaling up from imperfect detection models to estimate biodiversity measures – problem?

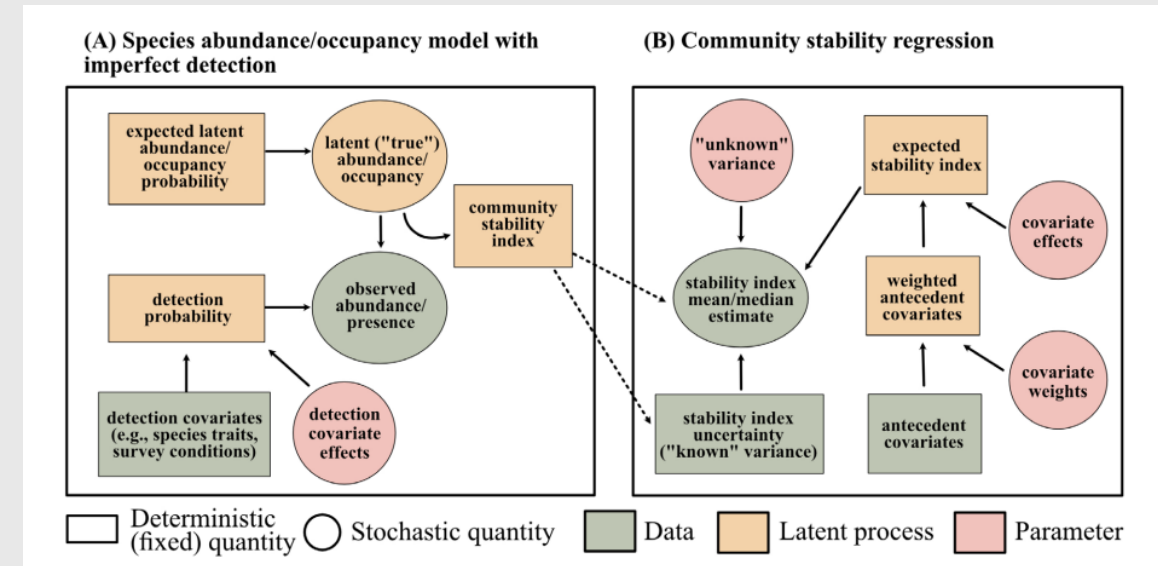


Quantify detection error? – Can do.

- Trying to correlate imperfect observation data to environmental data
- How accurate are your multispecies occupancy-abundance models?
- Scaling up from imperfect detection models to estimate biodiversity measures – problem?

Solutions

- Account for uncertainty in observed data
- Estimate a latent ('true') occupancy/abundance
- Model latent occupancy/abundance as a function of the environment
- Use derived values to calculate biodiversity and community metrics



Plant community data?

- Nested sampling designs
- Proportional data including 0 and 1
- Skewed distributions
- Too many zeros (zero inflated)



Plant community data? – You betcha

- Nested sampling designs
- Proportional data including 0 and 1
- Skewed distributions
- Too many zeros (zero inflated)

Solutions

- Hierarchical structure
- Two, linked models – presence/absence, cover.
- Cover model uses a Beta distribution that estimates latent cover.
- Also, quantify detection error.

Absence model

$$y_i \sim \text{Bernouli}(\psi_i),$$

$$\psi_i = \psi_i^C \times \psi_i^T + (1 - \psi_i^C),$$

$$\text{logit}(\psi_i^T) = \beta_0 + \sum_{v=1}^{12} \beta_v z_v + \sum_{v=2}^{12} \beta_{11+v} z_v \times z_1 + \eta_{s(i),j(i)} + \varepsilon_{j(i)}.$$

$$\text{logit}(\psi_i^C) = \alpha_0 + \alpha_1 \times \mu_i,$$

Models linked through latent proportional cover

Nested experimental design effects

Cover model

$$P(y_i^C | y_i^*, l) = \begin{cases} 1 & \text{if } l_{y_i^C} < y_i^* < l_{y_i^C} + 1 \\ 0 & \text{otherwise} \end{cases},$$

$$y_i^* \sim \text{Beta}(a_i, b_i)$$

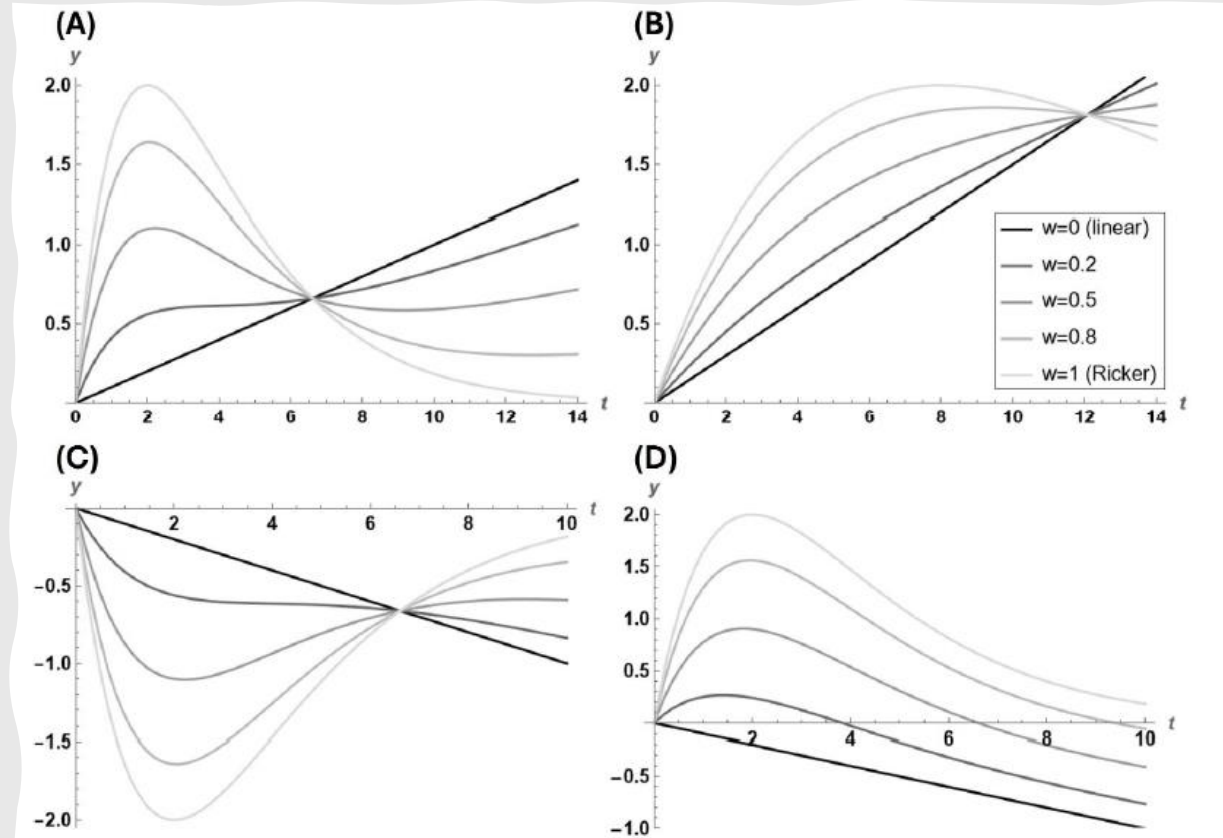
$$\mu_i = E(y_i^*)$$

$$\text{logit}(\mu_i) = \beta_0 + \sum_{v=1}^{12} \beta_v z_v + \sum_{v=2}^{12} \beta_{11+v} z_v \times z_1 + \eta_{s(i),j(i)} + \varepsilon_{j(i)}.$$



Characterize and compare response curves?

- Standard distributions not fitting well
- Compare responses across variables
 - Not all variables respond the same way

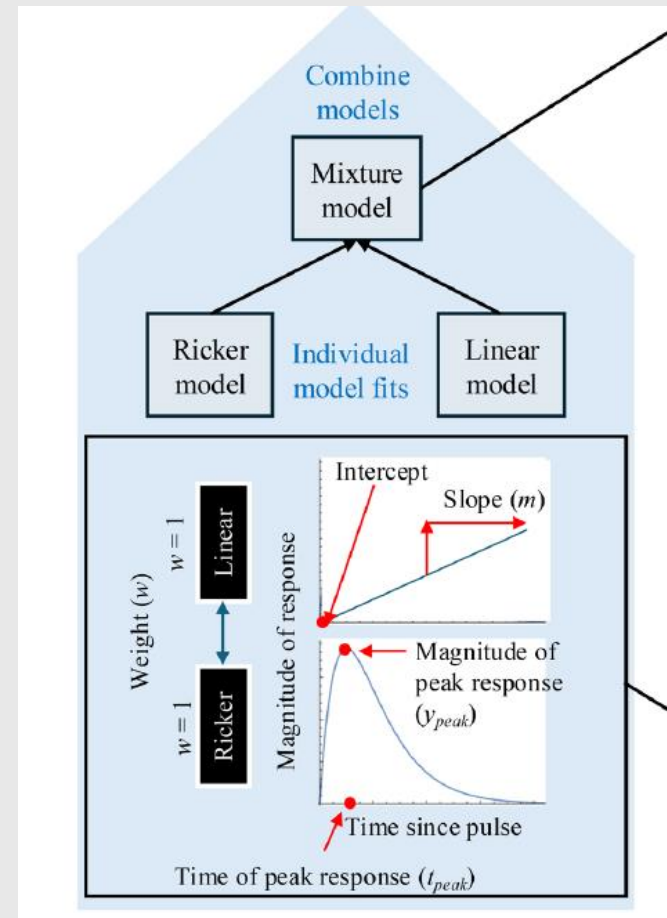


Characterize and compare response curves? – Of course.

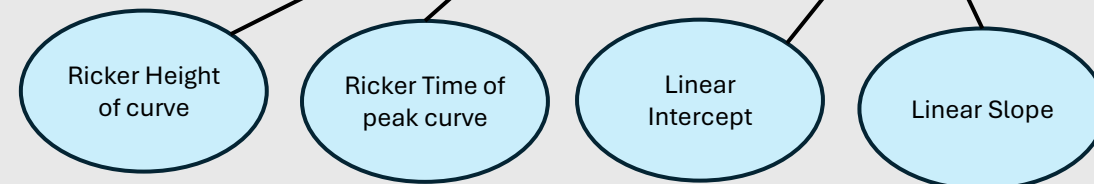
- Standard distributions not fitting well
- Compare responses across variables
 - Not all variables respond the same way

Solutions

- Use a more flexible distribution
 - Negative binomial vs. Poisson
- Estimate parameters that define the curve
- Model parameters in the same way, so that they are comparable

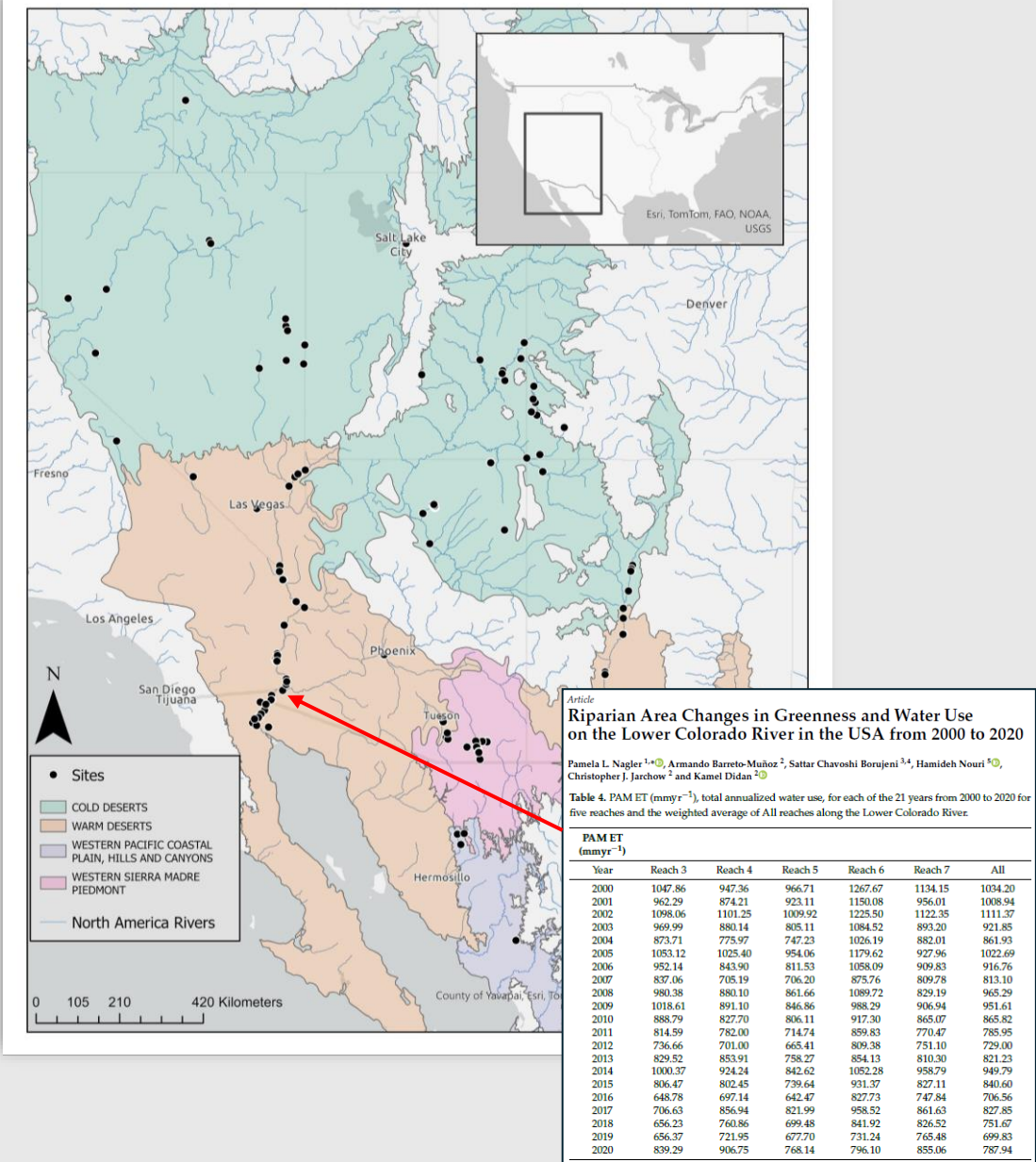


$$y(t, y_{peak}, t_{peak}, b, m, w) = w \cdot y_{Ricker}(t, y_{peak}, t_{peak}) + (1 - w) \cdot y_{Linear}(t, b, m)$$



Meta-analysis?

- Site, study-level covariates not consistently reported
- Differences in methods, variables, sample sizes
- Multiple values reported in studies
- Uncertainty associated with mean values



Meta-analysis? – Begs for Bayesian

- Site, study-level covariates not consistently reported
- Differences in methods, variables, sample sizes
- Multiple values reported in studies
- Uncertainty associated with mean values

Solutions

- Use all records, even if missing some data
- Means and variances can be inputs
- Hierarchical random and fixed effects
- Multivariate models

Log-response ratios and their variances are data!

$$\text{LRR}_i \sim \text{Normal}(\mu_i, \sigma_i^2)$$

$$\begin{aligned} \mu_i = & \beta_{v(i),s(i),t(i)} + \delta_{v(i),t(i)}^{\text{Dur}} \cdot \text{Dur}_i + \delta_{v(i),t(i)}^{\text{MAP}} \cdot \text{cMAP}_{s(i)} \\ & + \delta_{v(i),t(i)}^{\text{MAT}} \cdot \text{cMAT}_{s(i)} + \delta_{v(i),t(i)}^{\text{Int}} \cdot \text{cMAP}_{s(i)} \cdot \text{cMAT}_{s(i)} \end{aligned}$$

Multiple records for variables, studies, and treatment-types are accounted for.

$$\begin{pmatrix} \beta_{1,s,t} \\ \beta_{2,s,t} \\ \beta_{3,s,t} \end{pmatrix} \sim \text{Normal} \left(\begin{pmatrix} \beta_{1,t}^* \\ \beta_{2,t}^* \\ \beta_{3,t}^* \end{pmatrix}, \Sigma_t \right)$$

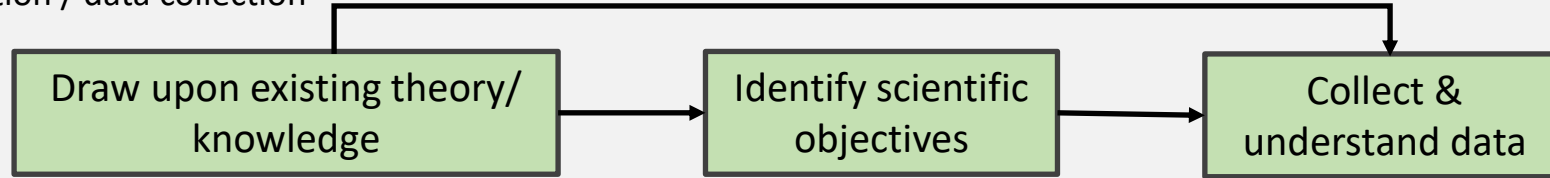
Responses are allowed to be correlated!

What you need to get started

- Basic understanding of probability concepts
- Familiarity with statistical distributions
- Fundamentals of calculus (e.g., integration)
- Good understanding of the study system and data

Modeling Process

Problem identification / data collection



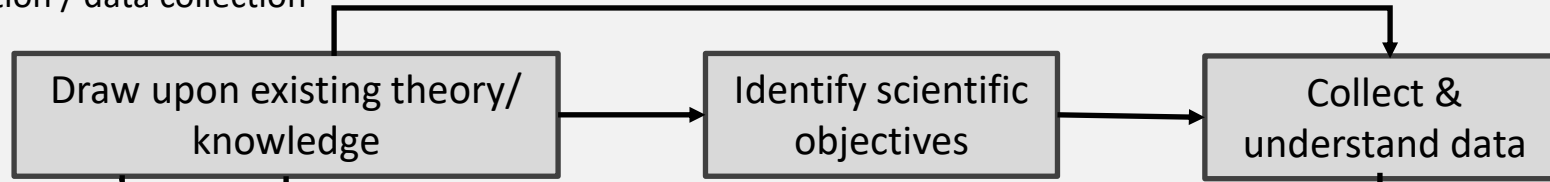
Model specification

Model
implementation

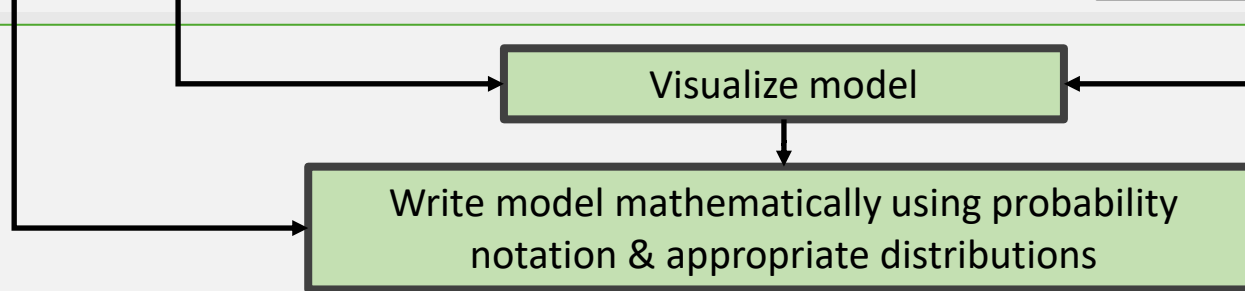
Model evaluation &
inference

Modeling Process

Problem identification / data collection



Model specification

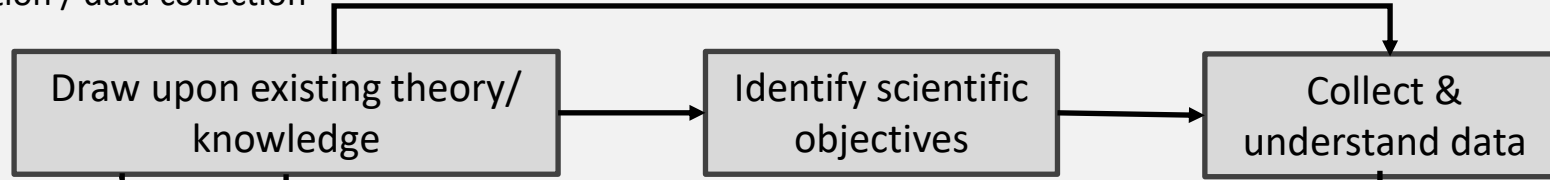


Model
implementation

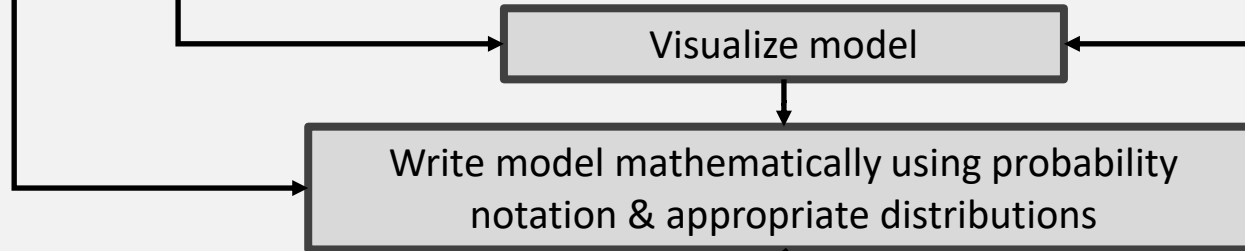
Model evaluation &
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Modeling Process

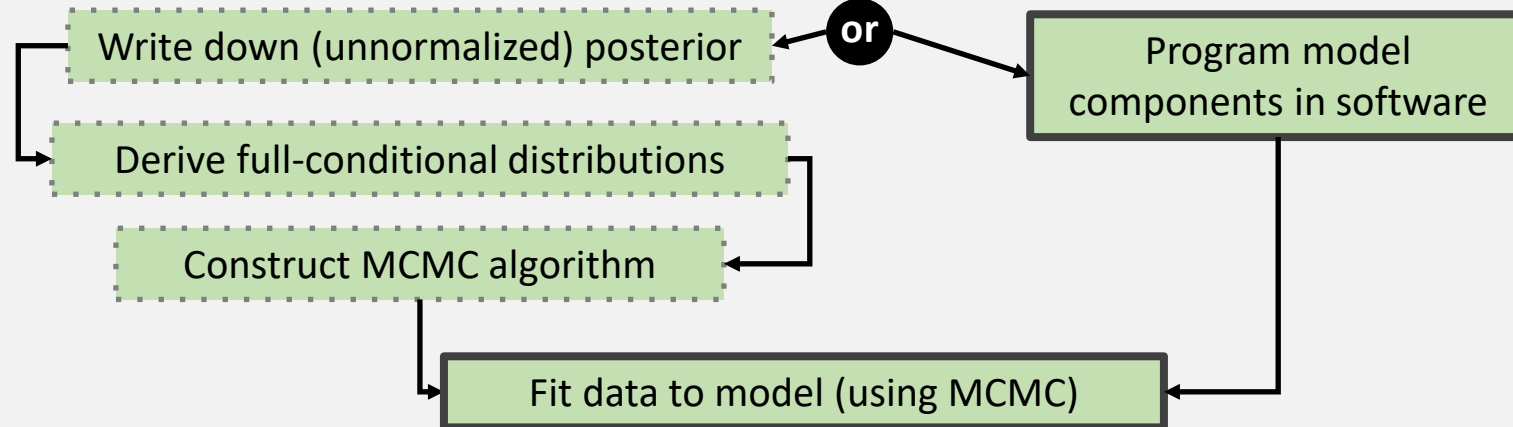
Problem identification / data collection



Model specification



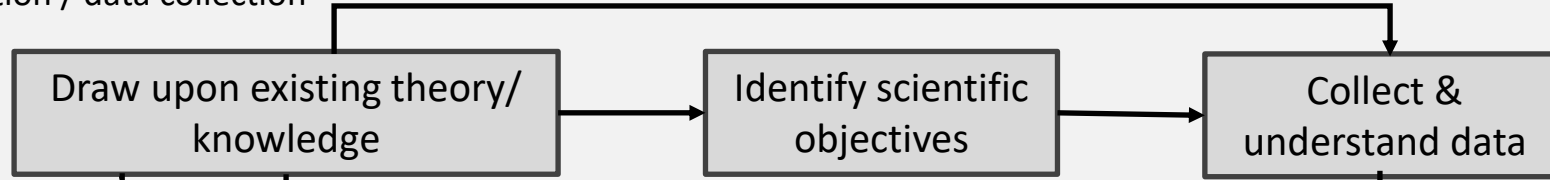
Model implementation



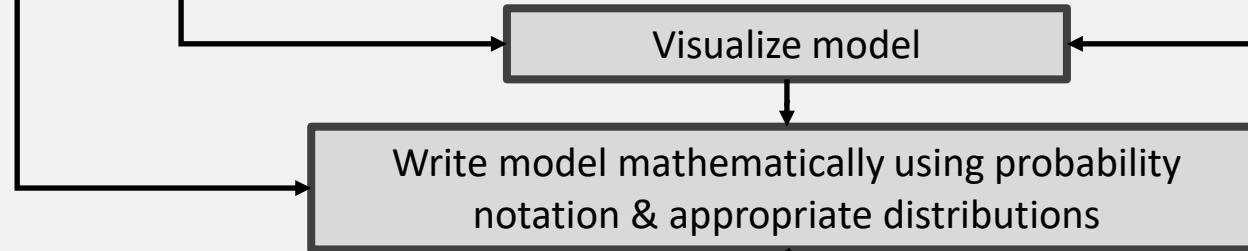
Model evaluation & inference

Modeling Process

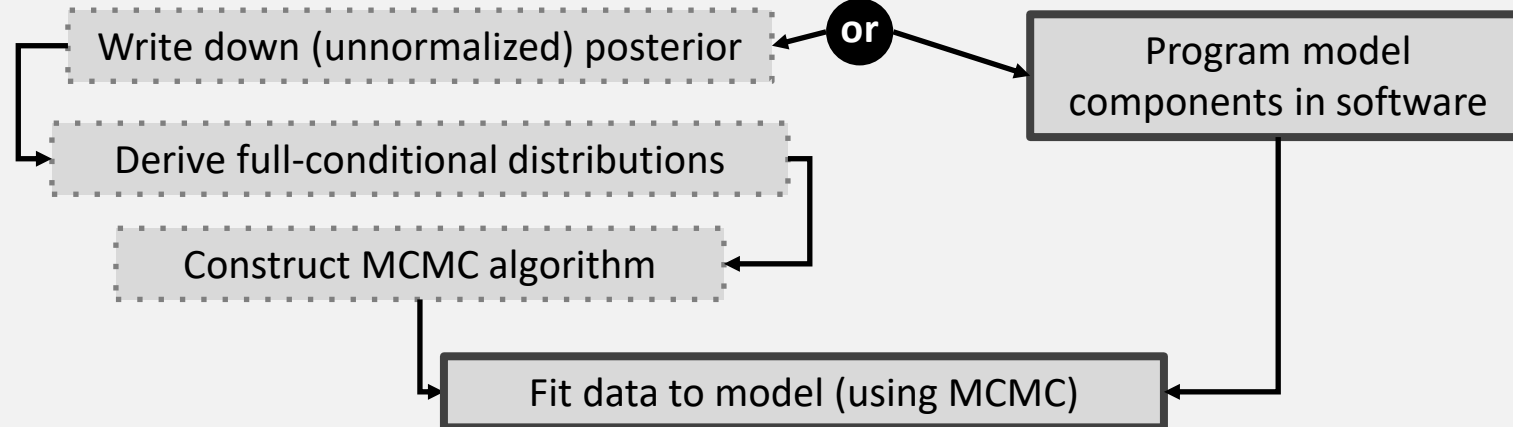
Problem identification / data collection



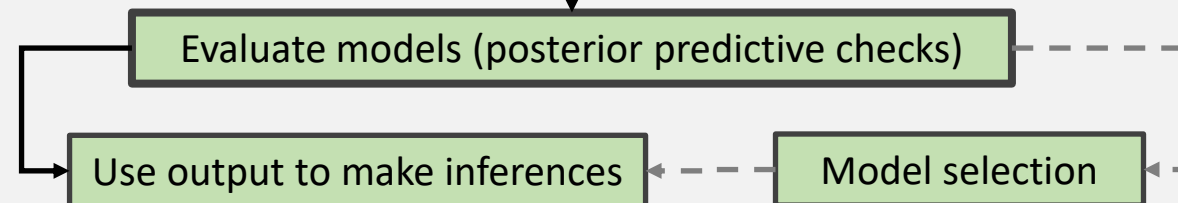
Model specification



Model implementation



Model evaluation & inference



Workshop Goals

- Understand fundamentals of Bayesian modeling
 - Statistical underpinning (Guo)
 - Fundamentals of hierarchical modeling (Ogle)
 - Basic coding tools and methods (Formanack)
- Implement basic hierarchical models (Working Groups)
 - JAGS

References Cited

- Miller-ter Kuile, A., Bui, A., Apigo, A., Lamm, S., Swan, M., Sanderlin, J.S., and Ogle, K., 2025, If You're Rare, Should I Care? How Imperfect Detection Changes Relationships Between Biodiversity and Global Change Drivers: *Global Change Biology*, v. 31, no. 7, p. e70362, <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.70362>.
- Ogle, K., Liu, Y., Vicca, S., and Bahn, M., 2021, A hierarchical, multivariate meta-analysis approach to synthesising global change experiments: *New Phytologist*, v. 231, no. 6, p. 2382-2394, <https://nph.onlinelibrary.wiley.com/doi/abs/10.1111/nph.17562>.
- Palmquist, E.C., Ogle, K., Whitham, T.G., Allan, G.J., Shafroth, P.B., and Butterfield, B.J., 2023, Provenance, genotype, and flooding influence growth and resource acquisition characteristics in a clonal, riparian shrub: *American Journal of Botany*, v. 110, no. 2, p. e16115, <https://bsapubs.onlinelibrary.wiley.com/doi/abs/10.1002/ajb2.16115>.
- Palmquist, E.C., Ogle, K., Butterfield, B.J., Whitham, T.G., Allan, G.J., and Shafroth, P.B., 2025, Hotter temperatures alter riparian plant outcomes under regulated river conditions: *Ecological Monographs*, v. 95, no. 1, p. e1645, <https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1002/ecm.1645>.
- Palmquist, E.C., Nagler, P., Ogle, K., DiMartini, C., Kennedy, J.R., and Sankey, J.B., 2026, JAMES BUTTLE REVIEW: A Synthesis of Riparian Plant Water Use Over Two Decades in North American Drylands: *Hydrological Processes*, v. 40, no. 2, p. e70408, <https://onlinelibrary.wiley.com/doi/abs/10.1002/hyp.70408>.
- Reich, E., Guo, J., Peltier, D., Palmquist, E., Samuels-Crow, K., Boone, R., and Ogle, K., 2025, Precipitation Pulse Dynamics Are Not Ubiquitous: A Global Meta-Analysis of Plant and Ecosystem Carbon- and Water-Related Pulse Responses: *Global Change Biology*, v. 31, no. 7, p. e70327, <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.70327>.